

SEMWEAVER: Provenance-Guided Refinement of LLM-Generated Vulnerability Checkers

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LLM-generated static vulnerability checkers can warn on both vulnerable and fixed code, or remain silent on vulnerable code. We present SEMWEAVER, an automated post-synthesis refinement workflow with patch-local, paired validation. Its editor receives provenance-labelled patch context, analyzer outputs, and analyzer-internal evidence: backend-produced representations and checker-execution observations bound to raw output and source scope. It retains warning-recovering or fixed-noise-reducing edits only after normal execution on both revisions.

We evaluate all 39 frozen, generated-only KNIGHTER CSA checkers: 15 initially warn on both revisions and 24 are silent on both. With the same GPT-6-Luna model, SEMWEAVER reaches 23 warning-positive vulnerable revisions versus 15 for KNIGHTER’s actual refinement loop, with patch-version $F_1 = 0.575$ versus 0.448. In the preexisting 15-case fixed-noisy stratum, SEMWEAVER retains vulnerable-side warning signals with 44 fixed-side reports versus KNIGHTER’s 69; the reduction is concentrated in one subject. Native versus no-internal refinement yields six comparative benefits, five disadvantages and nearly identical aggregate F_1 ; native’s additional coverage also increases overall fixed-side noise. Three paired decodes on 12 subjects show sampling-dependent effects. A separate automatic CodeQL example and three-model subset examine feasibility and configuration sensitivity. The results support conditional checker-completion benefits, not independently adjudicated target recall, universally superior precision or unseen-site generalization.

CCS Concepts: • **Software and its engineering** → **Automated static analysis**; • **Security and privacy** → **Systems security**.

Additional Key Words and Phrases: static analysis, vulnerability checkers, large language models, patch-guided refinement

1 Introduction

Static analysis is a cornerstone of software security assurance for systems code. Tools such as CodeQL [18] and Clang Static Analyzer (CSA) [37] detect vulnerabilities by checking source code against predefined rules without executing it [4, 7, 15, 48]. The effectiveness of these tools depends on the quality of their vulnerability checkers, each of which must encode the semantic conditions (path guards, state invariants, data-flow relations, and API contracts) that distinguish a real bug from safe code. Writing such checkers by hand requires deep expertise in both the target codebase and the analysis framework, making manual authoring expensive, slow, and difficult to scale.

LLM-based patch-driven synthesis offers a promising alternative. In this paradigm, a security patch serves as a compact before/after specification: the LLM consumes the patch and synthesizes a reusable checker that should fire on the vulnerable revision but remain silent on the fixed revision. KNIGHTER [57], a representative system, has demonstrated that this approach can produce compilable and executable CSA checkers from historical Linux kernel fixes.

However, the resulting checkers may be semantically inadequate. A security patch is inherently underspecified: a one-line goto-target change may encode an ownership barrier, a widened cast may encode a range guard, and a moved free may encode a lifetime invariant. A checker that captures only the syntactic shape of the edit, without encoding the underlying semantic condition, can fire on both the vulnerable and fixed revisions. After naïve refinement, the checker may instead lose the vulnerable-side trigger entirely, a failure mode we call over-pruning. These outcomes can

arise from incomplete API models, incorrect state updates, or a missing relation between the patch and the detector’s trigger.

Current refinement strategies, including KNighter’s own iterative loop, attempt to address this problem by operating at the output level [40, 52, 57]. They use execution reports and compiler feedback to guide further edits. KNighter’s report-driven loop does not attempt an edit when no report is available. The observed warning identifies a failed behavior, but may leave several plausible causes: a missing path guard, cleanup barrier, ownership transition, or API condition. An editor can reason from its patch and source context; additional analyzer representations provide another inspectable source of relevant structure.

We make a narrower observation: **some patch-relevant program structure is available in analyzer-produced intermediate representations** but is not ordinarily surfaced to the refinement loop. Path-sensitive analyzers such as CSA internally track branch guards, symbolic states, memory regions, and lifetime events during checker execution [37]. Relational query engines such as CodeQL maintain databases of data-flow edges, call-graph relations, type hierarchies, and API-usage patterns [2, 18]. These representations can suggest candidate predicates, but a CFG edge or call relation is not by itself a proof of path feasibility or checker correctness.

Exploiting these intermediate results, however, is not straightforward. Analyzer dumps and database relations can be much larger than the relevant patch scope. Selecting a bounded view also requires preserving its source binding and distinguishing backend-produced structure from diagnostic or source-derived context. The challenge is to select and present only the evidence that is relevant to the particular semantic gap in the checker under refinement.

SEMWEAVER provides post-synthesis checker editing guided by provenance-labelled analyzer context. SEMWEAVER is designed as a plug-in post-generation layer that is independent of the upstream LLM generator, treating any generated checker (from KNighter or another pipeline) as an artifact to be refined rather than regenerated. The framework addresses the evidence-selection challenge through three mechanisms. First, a rule-guided preflight extracts patch facts (e.g., guards, type widening, and affected calls) and prioritizes evidence requirements by inferred mechanism. The editor may request role views in the general workflow; the strict CSA comparison instead freezes its patch-scoped view before editing. For instance, a cleanup-edge patch triggers collection of path-guard and lifetime evidence from CSA, while a bounds patch triggers guard, flow, and API-usage evidence from CodeQL. Second, backend outputs, patch facts, source context, and validation feedback are normalized into a compact bundle with typed records, giving the editor a structured edit target (“add the patch-introduced cleanup barrier while preserving the freed-object transition”) instead of a vague behavioral symptom (“fixed side still warns”). Third, paired validation retains a candidate when it recovers a previously absent vulnerable-side warning, or reduces fixed-side reports while retaining vulnerable-side warning presence. Fixed-side silence is a stronger secondary outcome, not a prerequisite for every useful edit. This operational constraint does not independently establish target-vulnerability detection.

We evaluate all 39 frozen, generated-only KNighter CSA checkers. Fifteen initially warn on both revisions and 24 are silent on both; the initial zero PDS is a starting condition, not KNighter’s refinement result. All methods share starting artifacts and requested GPT-6-Luna high, with a 32-reply ceiling and their recorded stopping policies. Native SEMWEAVER reaches 23 warning-positive vulnerable revisions and version-level $F_1 = 0.575$, versus 15 and 0.448 for KNighter’s actual loop. The complete 15-case report-refinement stratum retains all vulnerable-side warning signals with 44 fixed-side reports versus KNighter’s 69, mainly through G24. Across all 39, native’s 123 fixed-side reports exceed KNighter’s 69 and no-internal’s 68. The native/no-internal comparison and

three paired decodes on the original 12 subjects therefore support conditional benefits and trade-offs rather than universal native superiority. Separate automatic CodeQL and fixed-subset model runs examine feasibility and configuration sensitivity.

This paper makes the following contributions:

- We distinguish output-level behavioral feedback from provenance-bound analyzer-internal evidence, and formulate paired-warning refinement for vulnerability checkers generated by LLMs.
- We present SEMWEAVER, a plug-in refinement layer that organizes provenance-labelled analyzer representations, analyzer outputs, patch and source context, and validation feedback into a compact semantic evidence bundle; the strict CSA comparison freezes the available view rather than evaluating adaptive routing.
- We provide an automatic, denominator-complete 39-checker comparison with KNighter’s actual loop; distinguish same-task fixed-side noise reduction from vulnerable-side warning recovery; and report a full-cohort ablation, repeated-subset variability, and cross-backend limitations without pooling incompatible evidence.

2 Background

2.1 Static Vulnerability Checkers

Static analysis examines source code without executing it, aiming to find bugs, security vulnerabilities, and code-quality issues at compile time or during code review [4, 15]. A vulnerability checker is a rule or query within a static-analysis framework that flags program locations matching a security-relevant condition. Modern frameworks support different checker paradigms. Path-sensitive engines such as the Clang Static Analyzer (CSA) perform symbolic exploration over program paths, tracking branch guards, symbolic values, memory regions, and lifetime transitions to detect bugs like use-after-free, null dereference, and resource leaks [37]. Relational query engines such as CodeQL compile source code into a queryable database and express checkers as declarative queries over data-flow edges, call-graph relations, type hierarchies, and API-usage patterns [2, 18]. Both paradigms have been deployed at industrial scale for continuous vulnerability detection [7, 48].

Writing effective checkers remains a bottleneck. A checker must encode not only the syntactic pattern of a vulnerability but also the semantic conditions (path guards, state invariants, data-flow relations, and API contracts) that distinguish real bugs from safe code. This requires deep domain knowledge of both the target codebase and the analysis framework, making manual checker authoring expensive and error-prone. Report burden also affects practical use [20, 23]; ranking and actionability prediction address emitted warnings [29, 47], whereas our refinement changes checker code.

2.2 LLM-Based Patch-Driven Checker Generation

Recent work has proposed using LLMs to automate checker generation from security patches. A security patch provides a compact before/after specification: the vulnerable revision should trigger the generated checker, and the fixed revision should not. This idea builds on a well-established line of work that treats patches as vulnerability specifications, including semantic patching [43], clone-based vulnerability discovery [22, 28], and patch-derived datasets [16]. KNighter [57] is a representative system that synthesizes CSA checkers from historical Linux kernel fixes and validates them against the vulnerable and fixed revisions.

Generated checkers may miss semantic conditions: a small patch can encode a guard, ownership transition, lifetime barrier or API contract beyond its syntactic shape. Existing refinement loops,

including KNighter’s own iterative strategy, operate at the output level [40, 52, 57]. They feed behavioral signals (e.g., “the fixed side still warns”) back to the LLM, revealing the symptom but not diagnosing the cause. Such feedback does not directly identify whether the missing predicate concerns a path guard, a cleanup barrier, an ownership transition, or an API contract; the model must infer the condition from the patch, source context and observed outputs.

2.3 Analyzer-Internal Intermediate Results

Static-analysis backends compute intermediate representations that can be exposed through debugging interfaces or database queries, but these are not normally supplied to a checker-refinement loop.

CSA, as a path-sensitive engine, internally maintains per-path symbolic states that record branch conditions, memory region bindings, allocation/free events, and object lifetime transitions [37]. CodeQL, as a relational engine, materializes a full program database from which data-flow edges, call-graph relations, type hierarchies, member-access patterns, and API-usage sequences can be queried [2, 18]. These intermediate facts have long underpinned taint analysis, graph-based detection, and interprocedural security reasoning [1, 25, 36, 55].

These representations may suggest a missing predicate, but a CFG branch or database tuple does not itself establish vulnerability semantics. The operational distinction between a backend-produced representation, a diagnostic, and a source-window fallback is central to our evidence accounting (Section 4.2). This motivates extracting bounded, provenance-labelled context for refinement while retaining normal paired execution and vulnerable-side warning presence as operational acceptance checks, not proofs of target-location correctness.

3 Motivating Example

This section traces an automatic refinement of the KNighter-derived CSA checker for Linux commit 97cba232549b, which repairs an off-by-one access in the AMD display driver [51, 57]. The example has an explicit safety relation: the largest look-ahead index must be smaller than the extent of the array actually accessed. Both the starting and refined checkers use `checkASTCodeBody`; the edit adds a missing bound relation to the existing analysis.

The bug and its patch. The member `dc->links` has declared capacity `MAX_PIPES * 2` (Figure 1, ❶). We abbreviate this constant as N in the figure; this is a presentation alias, not a new program variable. The function `get_host_router_total_dp_tunnel_bw` iterates over candidate links and, on a matching host-router branch, reads both `links[i]` and `links[i + 1]` (M10–M11, ❸). With the original condition $i < N$ (M5, ❷), the loop permits $i = N - 1$, so the look-ahead can reach index N , outside the array. The patch changes the bound to $i < N - 1$ (M6, ❹), which limits the largest look-ahead index to $N - 1$. Other branch conditions determine whether an access occurs, but the patched loop bound already prevents this final iteration from reaching an out-of-range look-ahead.

Why the baseline checker fails. The generated checker already recognizes an induction variable, a loop bound, and a look-ahead array subscript. It reports an access when it does not recognize a protective condition in an enclosing `if` (B1–B5, ❺). That rule overlooks a guard carried by the loop condition itself. The fixed program still contains `links[i + 1]`, so the checker reports on both revisions despite the reduced loop bound. The independent patch-related-object replay yields 1/1 alerts. The refinement problem is therefore to interpret the relation between the loop bound and array extent while retaining the unsafe access.

Evidence-guided refinement. The available bundle contains four provenance-labelled records: two strictly internal Clang 18 CFG/call-graph records and two analyzer-output records. The internal records anchor the affected loop and branch context to raw dumps from `debug.DumpCFG` and `debug.DumpCallGraph`; patch and source context identify the bound change. The declared array

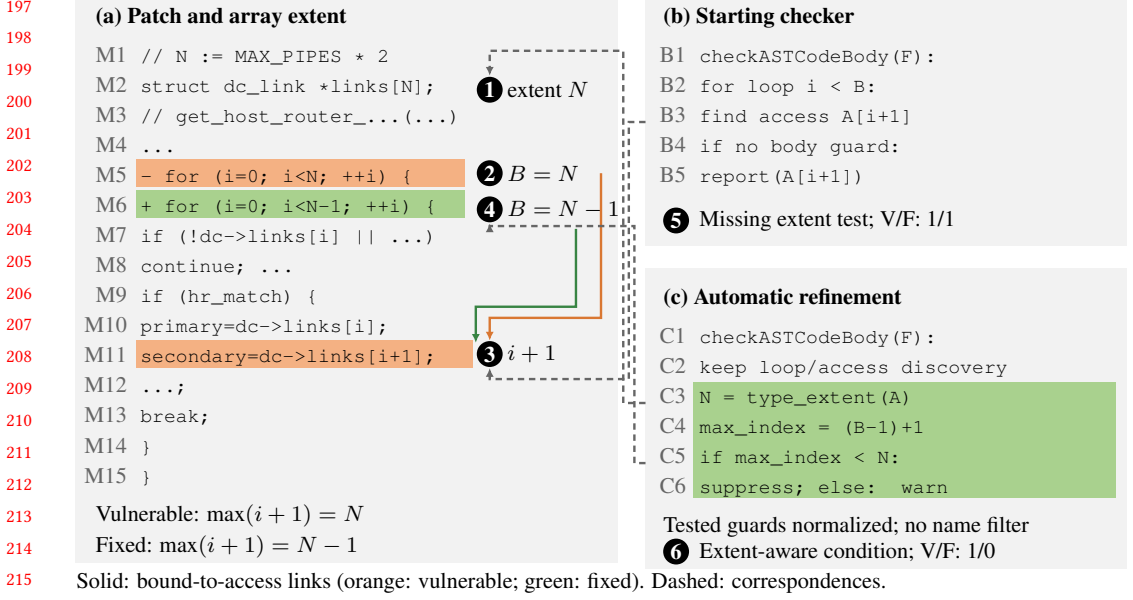


Fig. 1. **Capacity-aware look-ahead refinement.** Abridged code: N denotes $\text{MAX_PIPES} * 2$; hr_match denotes host-router equality. Original/refined vulnerable/fixed alerts are 1/1 and 1/0 in independent replay. The 14-response, model-produced illustration used fixture feedback and is separate from the one-response main-study checker. Orange/green solid arrows relate vulnerable/fixed bounds to the access; gray dashed arrows denote semantic correspondences. Code shading marks unsafe/patched bound and capacity logic.

extent is read from the AST type by the emitted checker. It is not a path-feasibility proof supplied by the evidence bundle.

The initial GLM-5.3-Flash edit adds a capacity-aware condition while retaining the original traversal. For the depicted loop $i < B$, the maximum look-ahead index is $(B - 1) + 1 = B$. A known extent N justifies suppressing the warning only when this index is smaller than N (C3–C6, ⑥). Consequently, the vulnerable $B = N$ remains reported and the fixed $B = N - 1$ becomes silent. This condition depends on type and index relations rather than the patched function name, a particular array capacity, or adjacency of the relevant statements. The initial edit uses two model calls and passes the independent Linux replay at 1/0.

Checking the example beyond its patch. Additional perturbation probes expose shortcomings in the initial candidate’s fallback and inherited guard recognition. Model-driven quality continuations receive these hash-bound diagnostics and repair the checker; the raw model edits and failed intermediate versions are retained. Four automatic GPT-6-Luna-high quality continuations use 12 additional model calls (14 including the initial Flash refinement). The final frozen checker retains 1/0 on the original Linux pair. It passes all 28 core fixtures, six boundary fixtures, 20 additional guard/bound fixtures, and 60 renamed/capacity-varied executions of those 20 forms: 58/58 unsafe and 56/56 safe executions overall. The labels for the latter 80 executions were also checked with AddressSanitizer [50]. These are finite experimental fixtures, not 114 independent bugs or an unseen-data confirmation set. The fixtures and diagnostic feedback were prepared through the research workflow with coding-assistant support; every scored checker-code edit is model-produced. The tests support robustness to the tested name changes, unrelated statements, constant capacities, and equivalent unit-step-loop/guard forms. Arbitrary aliasing, dynamic extents, and non-unit loop

updates remain outside this claim. This illustrative continuation is separate from the frozen 39-case results and does not establish a causal benefit from internal records alone. The one-response main-cohort G21 checker also reports 1/0, but passes only 74/114 of these same fixtures. The stronger 14-response illustration is not substituted into the 39-case scores.

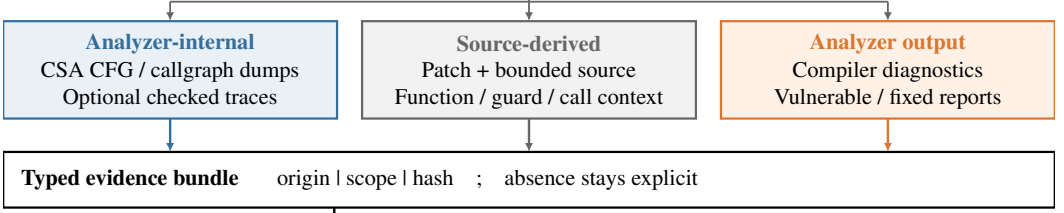
4 Approach

SEMWEAVER is a post-synthesis refinement layer for LLM-generated vulnerability detectors. Given a draft CSA checker or CodeQL query produced by an upstream generator, the patch that motivated the synthesis, and the corresponding source tree and validation scope, SEMWEAVER refines the detector to encode missing semantic predicates while retaining a vulnerable-side warning signal.

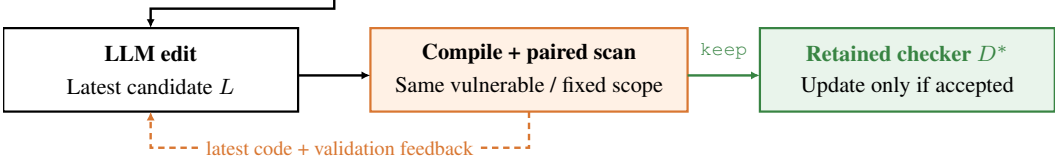
(a) Frozen inputs



(b) Evidence by origin



(c) Refine and retain



Gate: healthy + vulnerable warning; recover warnings or reduce fixed reports.

Native fixture guard is not a target oracle. L , D^* start at D ; all attempts are recorded.

Fig. 2. **SEMWEAVER architecture.** Blue denotes native evidence, gray source context, orange outputs/feedback, and green the retained checker; colors indicate roles, not correctness. Solid arrows carry inputs and candidates; the dashed lane returns latest code and feedback. Retention tracks warning presence and burden, not target correctness; fixed silence is secondary. All attempts remain auditable.

Figure 2 groups the workflow into three panels. **(a) Frozen inputs.** The input is an existing detector, patch, source tree and validation scope, not a new generation task. **(b) Assembly and bundle.** Rule-guided preflight, source extraction, analyzer collection and validation feedback supply provenance-typed records and bounded views (Sections 4.1–4.3). **(c) Paired-warning refinement.** The agent edits the latest attempted detector; normal paired outcomes update a separate best detector when they recover warning presence or reduce fixed reports while retaining it. Rejected attempts remain next-step input/feedback (Section 4.4). **Audit artifacts.** The verifier checks raw-output/source-scope bindings, interfaces, provenance counts, selected code and paired logs. Original prompts, replies and all attempts persist; an LLM explanation is not an origin certificate.

Algorithm 1 Paired-Warning Evidence-Guided Refinement**Require:** detector D , patch P , source S , scope V , backends A , budget K **Ensure:** refined detector D^* and audit report

```

1:  $R_0 \leftarrow \text{VALIDATE}(D, P, V)$ 
2:  $E_p, E_s \leftarrow \text{PREFLIGHT}(P), \text{EXTRACTSOURCE}(P, S, D)$ 
3:  $E_a \leftarrow \bigcup_{a \in A} \text{COLLECTNATIVE}(a, D, P, S)$     $E_v \leftarrow \text{FEEDBACK}(R_0)$ 
4:  $B \leftarrow \text{NORMALIZEMERGE}(E_p, E_s, E_a, E_v)$ 
5: if  $\text{PDS}(D, P, V)$  then return  $\text{AUDIT}(D, B)$ 
6:  $D^* \leftarrow D$     $L \leftarrow D$     $T \leftarrow \text{INITIALSTOPSTATE}$ 
7: for  $i = 1$  to  $K$  do
8:    $V_i \leftarrow \text{BOUNDEDVIEW}(B)$     $C_i \leftarrow \text{AGENTEDIT}(L, P, V_i)$ 
9:    $R_i \leftarrow \text{VALIDATEIFCOMPILABLE}(C_i, P, V)$ 
10:  if  $\text{ACCEPT}(C_i, D^*)$  then  $D^* \leftarrow C_i$ 
11:   $L \leftarrow C_i$     $E_i \leftarrow \text{FEEDBACK}(R_i)$ 
12:   $B \leftarrow \text{MERGE}(B, E_i)$ 
13:   $T \leftarrow \text{UPDATESTOPSTATE}(T, R_i, D^*)$ 
14:  if  $\text{STOP}(T, K)$  then break
15: end for
16: return  $\text{AUDIT}(D^*, B)$ 

```

▷ Equation 3

4.1 Evidence Selection and Exposure

The central design choice in evidence assembly (Figure 2b) is *what to collect and expose*. Analyzer outputs can be large and heterogeneous; passing them wholesale to the editor obscures patch-relevant relations. The preflight planner represents candidate requirements by role and priority, while source/function scope and bounded views limit what the editor sees. In the strict CSA comparison, collection is precomputed and the visible view is frozen, so its result does not measure demand-driven routing.

Rule-guided preflight and role requests. The implementation has no separately trained or prompted patch-mechanism classifier. PatchFactsExtractor parses the patch and metadata for affected functions, added guards, risky calls, and type widenings. EvidencePlanner takes a primary pattern from analysis metadata when available or infers it from those facts, then assigns fixed priorities to requirements: patch anchoring is 100, lifetime/state requirements for double-free or use-after-free are 92/89, and a guard for null or bounds cases is 90. It sorts by descending priority with evidence type as a tie-breaker and retains at most eight planned requirements in the default configuration. Table 1 summarizes mechanism-to-role examples; it is not a classifier confusion matrix or a promise that every role is analyzer-internal. The normalizer does not learn or re-rank facts. In the general workflow the editor may request evidence types, with duplicate and request-count limits. For the audited CSA comparison, static records are precomputed and additional requests disabled. The selected v43 implementation resolves compiler references into at most four function-local CFG witness cards, selected with patch/report scope; graph dominance is not path feasibility. It contrasts compiler/checker state associations and helper spelling with the detector, appending available execution observations. No-internal receives neither native witness nor dynamic views, but keeps patch, frozen source and ordinary feedback. This bounded deterministic exposure is not a classifier or a measured optimal routing policy.

4.2 Evidence Sources and Roles

Four collectors execute in sequence; Figure 2b groups views by origin rather than execution order. **Patch preflight** locates changed files/hunks and mechanism context. **Source extraction** resolves function, guard, call, state and build context in bounded views, without repository-scale dumps.

Table 1. Rule-guided evidence requirements by patch mechanism (representative); names are persisted record types, not gold labels or measured classifier outputs.

Patch mechanism	Possible record types	Backend
Cleanup edge / double free	allocation_lifecycle, state_transition, path_guard	CSA
Bounds widening / overflow	semantic_slice, dataflow_candidate, state_transition	CSA/CodeQL
Initialization / null deref	state_transition, path_guard	CSA
Use-after-free / escape	allocation_lifecycle, call_chain	CSA/CodeQL
API misuse / missing check	semantic_slice, call_chain	CodeQL

Analyzer evidence is classified by its actual origin, not by the collector’s name. We call a record *analyzer-internal* only if its payload is derived from a backend-produced intermediate representation, identifies the interface and raw output, and is bound to a source revision, file, and function. In the measured CSA cohort the interface is Clang 18’s `debug.DumpCFG` and `debug.DumpCallGraph`; the persisted UTF-8 dump is parsed into schema `semweaver.csa_cfg_snapshot.v1`, with a raw-output SHA-256 and source-scope binding. These dumps provide CFG branches and value bindings and call edges. They do *not* themselves prove a path-feasible lifetime or ownership transition. Analyzer diagnostics and alert counts are *analyzer output*, not internal state. Patch diffs, source windows, and compile-database-assisted source extraction are *source-derived* fallbacks even when a CSA or CodeQL collector emits them. A live CodeQL database query could meet the internal-evidence definition if its query, database revision, and tuples were retained; we do not attribute source-window-derived CodeQL facts to that category.

The selected implementation also observes instrumented CSA callbacks, abstract values/regions and checker-state associations. Raw JSONL uses `semweaver.checker_trace.v2`; bounded feedback uses `semweaver.trace_feedback.v1`. This is not concrete execution or exhaustive symbolic-state proof. Original/instrumented scans must preserve diagnostics on both revisions; caps, live fixtures and input hashes distinguish valid absence from broken instrumentation.

Extraction must handle verbose, backend-specific intermediates and instrumentation that may shadow identifiers, exceed caps or see no eligible callback. Revision, architecture, scope and parity checks retain missing observations as unavailable. Table 2 partitions 193 records across all 39 subjects. Its 81 native records comprise 45 guards, 27 state-transition and 9 allocation-lifecycle records; all 61 source-derived records are patch diffs, not source-window fallbacks. Attached context is not itself native. Dynamic availability is per-attempt (RQ3).

Table 2. Evidence primitives and provenance in the audited 39-case CSA cohort. One record may contain several CFG primitives; counts are by record origin, not primitive.

Origin	What the persisted record contains	Records
Analyzer-internal	Clang CFG guards, value bindings and call edges; raw-output and scope binding	81
Analyzer output	CSA diagnostics and paired validation reports; no internal state	51
Source-derived	Patch-diff records here; other interfaces may emit source-window fallbacks	61

Validation feedback records compilation, execution health and paired warnings: symptoms to diagnose, not semantic ground truth.

The seven persisted types are `patch_fact` (changed statements), `semantic_slice` (bounded statements, guards, API terms and symbols), `dataflow_candidate` (possible propagation), `call_chain` (call context), `path_guard` (control), `allocation_lifecycle` (resources), and `state_transition` (abstract state). “Barrier,” “type filter,” and “API usage” interpret record content; they are neither extra types nor proven path feasibility.

When a selected role yields no evidence, the bundle retains an explicit missing-evidence marker rather than treating the absence silently, so the refinement agent knows what was requested but unavailable.

4.3 Evidence Bundle Schema

The normalizer merges all collected facts into the `EvidenceBundle` shown in Figure 2b. Each fact becomes an `EvidenceRecord` carrying a type, the contributing analyzer, scope, source location, payload, provenance, and an optional `EvidenceSlice` (the compact semantic content consumed by the agent). The schema separates program facts from analysis rules in the spirit of relational analyzer frameworks [2, 5, 24], but uses a smaller role vocabulary tailored to checker refinement. Formally:

$$\begin{aligned}
 B &= \text{Norm}(E_p \cup E_s \cup E_a \cup E_v), \\
 E_a &= E_{\text{CSA}} \cup E_{\text{CodeQL}}, \\
 r \in B.\text{records} &= \langle \text{type}, \text{analyzer}, \text{scope}, \\
 &\quad \text{location}, \text{payload}, \text{provenance}, \text{slice} \rangle.
 \end{aligned} \tag{1}$$

Here E_p contains patch facts, E_s bounded source slices, E_a analyzer-internal representations or outputs filtered by selected roles (Section 4.1), and E_v validation outcomes. Each record’s provenance has one of three origins: `analyzer_internal`, `analyzer_output`, or `source_derived`. The normalizer deduplicates records and marks missing evidence; origin is neither a confidence score nor a proof of semantic correctness.

The type-to-backend mapping reflects available interfaces: CSA CFG/call-graph dumps inform guard, state, and call-chain records; CodeQL queries can support flow and relation records when a live database is available. These are record types, not seven distinct backend analyses. Cross-function call edges provide context, but the present CSA dump does not establish an interprocedural path or ownership proof. Provenance remains visible when a type is populated only by a fallback.

The bundle is bounded by design: scope and selected views limit raw dumps. The matched pipeline contrast and repeats remain subject to selection history, stochastic draws and bundled native components; they do not isolate each representation’s causal effect.

4.4 Paired-Warning Refinement

The refinement loop (Figure 2c) receives the current detector, patch, baseline validation notes, and a bounded role-specific evidence view. The interface supports requesting further roles from the persisted bundle, but the provenance-audited matched condition freezes the visible view before the edit and disables additional agent requests. Thus its measured effect is a contrast between preselected native-evidence and no-internal views, not a test of dynamic evidence retrieval. This keeps the prompt bounded and the two conditions auditable.

The agent’s tool-mediated loop draws on recent LLM agent and self-refinement architectures but constrains actions to attached evidence, detector edits, and validation commands [40, 49, 52, 59]. Where evidence requests are enabled, they replay attached records rather than launch open-ended repository exploration; the request count is capped and unused or missing evidence is logged. In the matched condition, request count is zero by protocol.

Model versus deterministic components. LLMs may produce patch-analysis metadata and the editor’s checker edit; when enabled, the editor also chooses which evidence types to request. The preflight requirement rules, raw-output hashing and scope binding, record normalization, checker compilation, paired replay, and acceptance are programmatic. The LLM does not create a backend dump, set a provenance origin, or decide PDS. The audit retains the exact starting checker, prompt and response, visible roles, raw evidence, candidate source, and validation decision. This boundary prevents an LLM explanation from being counted as analyzer-produced evidence. The editor prompt supplies the current and starting checker paths, patch path, bounded source/evidence view, baseline vulnerable/fixed symptom, backend identity, and edit/validation rules. It asks for an in-place checker edit that models the mechanism rather than matching patch spellings, forbids suppressing the vulnerable trigger, and requires compilation before paired acceptance. Compiler-repair prompts receive the actual build diagnostics; all available English templates and rendered exchanges are named in the artifact manifest, with absent historical raw exchanges explicitly marked rather than reconstructed.

Retention gate. Changed candidates must compile and receive normal paired execution. Equation 3 retains warning recovery from silence, or lower fixed reports with a nonzero vulnerable signal. PDS is stronger but not mandatory. The native-only controlled-fixture guard preserves required existing hits; it is not a target oracle. Silencing an existing vulnerable signal is rejected. A comparative disadvantage against control may instead reflect failure to recover from initial silence or a smaller improvement.

Feedback loop. The next permitted reply edits the latest attempt using its build or paired feedback, while the best retained detector is separate. Rejected and invalid edits remain visible. Positive/fixed-silent saturation, three valid non-improving validations, three consecutive invalid attempts or the reply cap stops the case (Algorithm 1). Execution interruptions are unscored and repaired without resampling consumed prefixes. The 15-case report-refinement view uses the same full-cohort outputs; it is not a separate per-case-budget experiment.

Backend extension surface. An adapter must provide registration, evidence/provenance mapping, dependencies, execution/result decoding, and paired validation. Shared modules still contain backend branches. Four enumerated CodeQL-specific modules occupy 2,438 physical lines, including comments and generation code: an existing footprint, not code to rewrite for each backend. With a usable analyzer and patch pair, we estimate that a familiar developer using Codex can complete basic adaptation within an afternoon by reusing these interfaces. This is a planning estimate, not measured labor or a scalability result. New native-evidence semantics and broad validation require separate work. The artifact lists prerequisites and tasks. A new bug mechanism needs a role mapping and safe/unsafe fixtures, and a new extractor only for unavailable relations.

5 Evaluation

We evaluate automated patch-related warning recovery and report reduction, not whole-kernel bug discovery. Checker outcomes and execution health are recorded separately.

5.1 Research Questions

RQ1: Generated CSA checkers. From identical starting artifacts, how do KNighter’s actual loop and SEMWEAVER affect vulnerable-version warning coverage, fixed-side report burden and paired discrimination?

RQ2: Cross-backend feasibility and cost. Can a separately archived automatic CodeQL query be refined and independently replayed, and what integration surface and execution costs are observable?

RQ3: Evidence sensitivity. How do native-visible and no-internal pipelines differ on all 39 subjects and three paired decodes of the original 12-subject subset?

RQ4: Model configuration sensitivity. What outcomes arise on those 12 subjects under three editor configurations with common reply/output ceilings?

5.2 Subjects

E1 freezes one automatically generated CSA checker per selected Linux commit from KNighter’s upstream artifacts. The inventory contains 286 generated records on 60 commits; 241 records on 39 commits have no upstream refinement-status row. A frozen rule selects one per commit by upstream TP, then TN, then lowest checker index, taking the nonempty repaired checker source. This non-random ranking defines inputs only: independent paired replay supplies our outcomes. No subject is removed because refinement fails or is noisy. Initially 15 checkers warn on both revisions and 24 are silent on both; zero initial PDS is a starting state, not KNighter’s refinement result. No manual checker completion enters scored portfolios.

All39 exercise actual KNighter refinement and native/no-internal SEMWEAVER. The 15 initially fixed-noisy subjects form the complete preexisting report-refinement stratum, not a favorable result-selected subset. On the other 24, KNighter’s actual no-report branch compiles but makes no model edit; SEMWEAVER attempts warning recovery. E1 is a complete system comparison, not evidence that KNighter tried and failed 24 recovery edits.

The selected whole implementation is v43, chosen after comparing two configurations on six experimental subjects. Exact compatible executions on those subjects are reused in E1; selection probes are not an independent confirmation set. Source revisions, reuse equivalence and earlier trials remain available. G04 uses the architecture-correct ARM64 scope and evidence, not a zero substituted for earlier wrong-object execution.

E2 is a separate automatic CodeQL example from an archived independent generator, with three decodes of the same input. Historical mixed/manual 20-row claims are withdrawn. CodeQL is neither pooled into CSA 39 nor compared against KNighter. E3 includes all 39 main pairs and the original 12 subjects with three decodes per arm: first from E1, then 48 additional cells. E4 uses that same 12 across three native editor configurations, one decode each, 36 cells. The subset was fixed before this final round and lacks buffer-overflow/out-of-bound categories. Repeated decodes are not independent bugs; E4 is not full 39 or multi-run cross-model robustness.

There is no separate LLM patch-mechanism classifier to score. Fixed rules prioritize evidence requirements; the strict comparison preloads its native view and disables new role requests. We claim neither classifier accuracy nor optimal routing.

5.3 Metrics

Primary outcomes are vulnerable-version warning presence and fixed-side report burden. Let $n_v(D)$ and $n_f(D)$ count warnings on the same vulnerable/fixed patch-related object scope. The stronger secondary outcome PDS is:

$$\begin{aligned} H_v(D, P, V) &= \#[\text{alerts}(D, V_{\text{vuln}}, P) > 0], \\ S_f(D, P, V) &= \#[\text{alerts}(D, V_{\text{fix}}, P) = 0], \\ \text{PDS}(D, P, V) &= H_v(D, P, V) \wedge S_f(D, P, V). \end{aligned} \tag{2}$$

PDS is not required for every useful edit. Recovery from silence may retain fixed noise; a warning-positive checker may improve by reducing fixed reports (Table 3).

Table 3. Measured warning outcomes, not adjudicated target labels.

Term	Operational meaning
Warning recovery	Initial vulnerable silence becomes warning-positive; fixed noise remains visible.
Report reduction	Fixed reports decrease while vulnerable warning presence remains.
PDS	Vulnerable warning-positive and fixed silent; a stronger secondary outcome.
Signal loss	A previously warning-positive vulnerable revision becomes silent; rejected.
Comparative disadvantage	Another arm obtains a better retained outcome; not necessarily own-start regression.

For retained detector D^* and normally paired-executed candidate C , the improvement rule is:

$$\begin{aligned} \text{Accept}(C, D^*) = & \text{AnalyzerOK}(C) \wedge H_v(C) \\ & \wedge (\neg H_v(D^*) \vee n_f(C) < n_f(D^*)) \wedge Q(C). \end{aligned} \quad (3)$$

The editor’s local gate requires a changed artifact before paired assessment. Q preserves required hits on controlled array fixtures in native; it is identically true for no-internal. This native-only guard records zero rejections in E1/E3, but its overhead and policy asymmetry remain disclosed. It is not a target-location oracle.

We binarize each checker/revision by warning presence. Each vulnerable revision is one positive and its fixed revision one negative; precision, recall and $F_1 = 2TP/(2TP + FP + FN)$ describe 78 patch-version decisions, not individual bug sites. Fixed reports measure inspection burden, not confirmed false-positive bugs. G08 warns on allocation failure rather than the registration-failure leak repaired by its patch; G24’s remaining warnings have unverified target alignment. No exhaustive target-recall claim follows. Failed scans are unscored, never substituted as zero warnings.

5.4 Experimental Environment

CSA uses pinned Clang/LLVM 18.1.3 and Python 3.12.3; CodeQL uses 2.26.4, codeql/cpp-all 8.0.3 and no-build databases. Each Linux subject uses its patch commit and parent with allyesconfig, architecture-correct objects and identical starting/baseline/candidate scope. Manifests bind code, evidence and raw paired outputs. Requested model labels and high effort are retained; service labels do not establish immutable revisions. Configured temperature 0 is not enforced sampling control: both Responses adapters omit it. We claim no fixed seed or deterministic decoding.

Generative AI in research. LLMs generated starting checkers and edit candidates. Extraction, provenance checks, compilation, paired scans and numerical outcomes are programmatic. Codex assisted with drivers, audit/analysis code and perturbation preparation; it supplies no ground-truth target labels. Original messages, English reading companions, candidates and script hashes are retained; authors verify AI-assisted code and interpretations.

5.5 Baselines and Variants

E1 native, no-internal and KNighter share starting checkers, requested GPT-6-Luna high and a 32-reply ceiling. KNighter retains its five-report/two-outer-attempt/report-driven algorithm; its actual emitted artifacts receive independent scans. Its 24 no-report branches consume no reply. Two natural one-response No-FP stops are equivalence-audited reuses with their original eight-reply ceiling retained, not relabelled as fresh32-reply runs; termination occurs before either ceiling. The 15-case view uses these same outputs, not a favorable per-case allocation. Equal maximum replies do not mean equal actual calls, tokens or CPU work.

Within SEMWEAVER, no-internal withholds native compiler and checker-execution views while keeping checker, patch, frozen source context, build/paired feedback and stopping rule. Native

receives bounded CFG witness/mismatch views and available dynamics; absence is explicit. E3 repeats the original 12 at 32 replies and 16,384 output tokens per call. E4 evaluates GPT-6-Luna high, GLM-5.3 high and GLM-5.3-Flash high with 2 replies and 65,536 output tokens per call. Actual GLM effort fields are retained. E4 is limited-budget configuration sensitivity, not equal-compute model ranking or cross-model native/control causality. Earlier default-effort, smaller-output and outside-subset histories stay separate.

5.6 Protocol

Each permitted reply produces a tool-mediated edit; compilable changed candidates receive independent paired replay. The next call starts from the latest attempted artifact and its failure/paired feedback, while the best retained detector is separate. Normal candidates follow Equation 3. Stopping occurs at positive/fixed-silent saturation, three valid validations without improvement, three consecutive invalid model attempts, or the reply cap. Rejected edits, compile/format failures, observer unavailability, provider interruptions and invalid scans remain distinct.

Repairs preserve original failures and consumed prefixes. Thinking-only received replies are charged, not resent free. One GLM reply contained an explicit action and three edits but omitted explanatory summary; network-disabled replay supplied only an empty summary, reproduced exact model-specified code and made zero new calls. Healthy zero-callback probes use static evidence without invented traces. Repairs do not authorize manual scored-code edits, changed budgets or failed scans counted as negatives.

6 Results

6.1 RQ1: Generated CSA Checkers

Table 4 reports all 39 subjects and the preexisting 15-case noisy stratum. Zero starting PDS is an entry condition, not KNighter’s refinement result. Every portfolio retains the 15 initially warning-positive vulnerable revisions. Among the 24 initially silent subjects, native recovers 8, no-internal 6, and KNighter 0 under its actual no-report policy. These are warning signals, not verified target-recall counts.

Table 4. CSA comparison: (a) all 39 subjects; (b) the complete preexisting 15-case fixed-noisy stratum, reusing the same outputs. TP/FP/FN are version-warning decisions; fixed reports measure burden and replies count received responses. Color shades 1 – F_1 at .25/.50/.75, not significance.

(a) Full cohort: 39 subjects

Condition	Version outcomes				F_1	Burden / effort	
	PDS	TP	FP	FN		Fixed reports	Replies
Unmodified	0	15	15	24	.435	73	0
KNighter loop	2	15	13	24	.448	69	179
SEMWEAVER, native	5	23	18	16	.5750	123	150
SEMWEAVER, no internal	8	21	13	18	.5753	68	133

(b) Noisy 15: all methods retain TP=15, PDS=2 and $F_1 = .6977$

	KNighter loop	Native	No internal
Fixed reports	69	44	68

The additional coverage has a noise cost: native’s 123 fixed reports exceed no-internal’s 68 and KNighter’s 69; its precision is 0.561 versus 0.618 for no-internal. Nearly tied F_1 does not establish native superiority. G12 contributes 73 distinct fixed-side warning locations, not duplicates; its

broader matching yields 74 vulnerable and 73 fixed reports. This unfavorable tradeoff remains in the aggregate.

In the complete 15-case fixed-noisy stratum (Table 4b), native reduces reports by 25 (36.2%) against KNighter and 24 (35.3%) against control. The effect is concentrated in G24’s 27 $\rightarrow$ 2 fixed reports; other small differences offset one another. Binary fixed positivity and F_1 are identical across methods. G24’s target preservation is unverified, so lower burden does not guarantee sound suppression.

6.2 RQ2: Cross-Backend Feasibility

A separate archived automatic ImageMagick CodeQL query starts at 0/0 vulnerable/fixed rows. Table 5 includes two 8/0 recoveries and the third decode’s 22/22 outcome. The successful edit corrects relative-path matching while keeping exact file/function/local-variable filters. This demonstrates patch-local operation on a second format, not broad backend transfer or native-evidence causality. A second query already starts at 2/0 and is not counted as a gain. KNighter is not a CodeQL baseline.

Table 5. Three automatic CodeQL decodes of one query. Counts are vulnerable/fixed output rows; time is recorded agent-round duration, excluding separate paired replay.

Decode	V/F rows	Replies	Tokens	Agent time (s)
1	8/0	1	15,122	87.87
2	8/0	1	14,345	42.74
3	22/22	8	110,282	516.77

Second-backend costs. Table 5 reports the original model runs. Three warm paired replays per query preserve 0/0 and 8/0. Median query-plus-decode time is 12.57 s initially and 36.51 s after refinement with two threads; peak process RSS is 976/1,078 MiB. Database-creation log intervals span 31/30 s, excluding unlogged startup. Developer labor is unmeasured. These one-case existing-environment observations are neither a speedup, cold-install benchmark nor whole-project scalability estimate.

6.3 RQ3: Internal-Evidence Sensitivity

Table 2 reports initial record provenance; Section 4.2 defines native types and source-derived context. Dynamic availability has a separate denominator: 103 of 150 unique native-chain attempts have usable hash-bound observations; 11 are unavailable due to candidate compilation, 18 unsupported context, 13 zero scoped callbacks, 1 reproduced candidate crash and 4 capped optional observations. None establishes safety from absence.

The full 39 comparison has six native comparative benefits, five disadvantages and 28 ties/other outcomes. Changes include cleanup-wrapper recognition in G19, nested dereference/binding recovery in G25, and null/region/frame handling in G24. G19 adds `__free_kfree` to a cleanup predicate, yielding 15/0 versus control’s 0/0; 15 reports are not 15 bugs. Conversely, control recovers symbolic-region handling in G23 and persistent-field release recognition in G38 without native records. G12 drops CPU-role restrictions, explaining its broad matching and noise. Code differences are observable, but a record’s presence or an LLM explanation alone does not prove edit causality.

Figure 3 shows five subjects tying in all three decodes and seven with mixed directions; none benefits comparatively in all three. G19’s native 15/0 recurs, but control catches up in its third decode. G24’s main reduction does not repeat. G14’s 3/3 $\rightarrow$ 2/2 improves its own start but trails

Subject	Decode 1 (E1)	Decode 2	Decode 3
G04	= 0/0 : 0/0	= 0/0 : 0/0	= 0/0 : 0/0
G08	= 1/1 : 1/1	− 2/2 : 1/1	= 1/1 : 1/1
G09	+ 1/1 : 0/0	= 1/1 : 1/1	= 1/1 : 1/1
G13	= 0/0 : 0/0	= 0/0 : 0/0	= 0/0 : 0/0
G14	− 2/2 : 2/1	= 2/1 : 2/1	= 2/1 : 2/1
G15	= 9/9 : 9/9	= 9/9 : 9/9	= 9/9 : 9/9
G19	+ 15/0 : 0/0	+ 15/0 : 0/0	= 15/0 : 15/0
G24	+ 2/2 : 27/27	= 3/3 : 3/3	= 27/27 : 27/27
G25	+ 2/2 : 0/0	= 0/0 : 0/0	= 0/0 : 0/0
G36	= 1/1 : 1/1	= 1/1 : 1/1	= 1/1 : 1/1
G37	= 0/0 : 0/0	= 0/0 : 0/0	= 0/0 : 0/0
G38	− 0/0 : 1/0	= 1/0 : 1/0	= 1/0 : 1/0

Native V/F : no-internal V/F. + benefit; − disadvantage; = tie/other.

Recovery +4 / −1; fixed-report change +1 / −2; tie/other 28.

Fig. 3. E3: all 36 paired decodes on 12 subjects; cells compare native and no-internal V/F counts. Decode 1 reuses E1; the remaining decodes add 48 condition cells. Color and sign encode the comparative direction.

control’s 2/1; later decodes tie. In the main comparison, G05/G23/G38 retain initially 0/0 in native while control recovers warnings, so their disadvantage is not loss of an initial signal. Repeated subjects, selection/reuse and native-only gating limit stronger attribution. Realized call differences under a shared stopping rule are outcomes/mediators, not independently assigned confounds.

6.4 RQ4: Model Configuration Sensitivity

Table 6 uses the original 12 subjects, native only, one decode per model, 2-reply and 65,536-output ceilings with requested high effort. All configurations retain the five initial vulnerable warning signals and recover additional versions. Flash has the strongest observed outcome here, but one decode on this subset does not establish general ranking or variance-controlled model robustness.

Table 6. Limited-budget native sensitivity: 12 subjects per model, not full-39 robustness. Computation differs despite common reply/output ceilings. F_1 shading follows Table 4.

Editor	TP	FP	PDS	F_1	Fixed reports	Replies
Unmodified	5	5	0	.455	42	0
GPT-6-Luna high	7	6	1	.560	40	24
GLM-5.3 high	7	5	2	.583	16	23
GLM-5.3-Flash high	9	5	4	.692	11	21

Received replies total 24/23/21 respectively. Capped, format-invalid and compile-failed replies remain charged; a healthy previous detector is retained when no valid improvement occurs. These are complete workflow outcomes, not successful code generation on every call. No no-internal arm is run across models, so E4 cannot isolate cross-model evidence effects.

6.5 Qualitative Findings

The automatic motivating checker retains AST traversal and adds actual-capacity reasoning, with 14 received replies and fixture-guided corrections. Its 114 finite successes are separate from the main one-response checker’s 74/114. The case ledger retains all 11 non-tied main comparisons,

selected diffs and intermediate outcomes. Three positive selected candidates have usable dynamics (G19/G24/G25), and three use static records with unavailable dynamics (G09/G12/G35). All six gains therefore cannot be credited to dynamic traces. The findings support conditional checker completion and auditable tradeoffs, not universal precision improvement, proof of target preservation or unseen-family recall.

7 Discussion

Why post-synthesis and LLM-generated checkers? Holding the checker fixed attributes improvement to an edit rather than a new generation draw. The interface can also accept a compatible handwritten checker and paired target, although that population and evidence-assisted fresh generation remain unevaluated.

Evidence as context compression. Provenance-tagged, patch-scoped views bound the editor’s context instead of exposing full repository and analyzer dumps. This frames refinement as specification completion rather than report suppression [12, 39, 41, 42, 53]; the present study does not isolate compression efficacy.

Backend complementarity. The CSA study and separate E2 CodeQL replay exercise two detector formats. They do not show that every evidence role was populated from an analyzer-internal representation. In the audited CSA cohort, the verified native interface is a CFG/call-graph dump; source-derived and diagnostic records remain separately labelled. A CodeQL-native coverage claim requires retained database-query provenance.

Failure modes. Over-pruning can silence a retained vulnerable signal and is rejected. Broader matching can instead recover warnings with substantial fixed noise, as in G12; the retention rule permits that tradeoff, which must remain visible. Other failures include missing observations, unsupported interfaces, invalid model output and compilation errors. A healthy retained checker is not replaced by a fabricated zero-warning execution.

Limited semantic generalization. SEMWEAVER studies patch-local warning behavior [8, 26, 44, 45]. The separately budgeted motivating checker uses extent/index relations rather than patch names or adjacency and passes finite perturbations of names, statements, capacities and unit-step-loop/guard forms. Diagnostic feedback informed its revisions; these are not unseen-family recall estimates. The main one-response checker passes fewer fixtures. Arbitrary aliasing and dynamic extents remain outside scope.

Precision and scaling boundary. For patch-related checking, the complete 15-case stratum retains all vulnerable warning signals with 44 fixed reports versus KNighter’s 69, mainly through G24. Across all 39, native recovers eight initially silent subjects but produces 123 fixed reports, exceeding both comparators; no-internal nearly ties its F_1 . Static evidence covers all 39; dynamic availability varies by attempt. This supports automated bounded processing and conditional completion, not uniformly improved precision, whole-kernel throughput or target recall. Changed forks or later versions require stability tests and separate target-location checks. Each model configuration reduces fixed reports while increasing coverage on 12 subjects, without cross-model repetitions. Warm CodeQL costs, implementation footprint and the conditional afternoon adaptation estimate answer distinct questions, not combined measured labor or scalability.

8 Threats to Validity

Internal. Collection, editing or validation errors can create spurious outcomes. Candidate compile/format failures are distinct from invalid scans; scanner crashes remain unscored with replies charged and are repaired without resetting prefixes. Zero-callback/capped observations remain unavailable, not safe. Raw hashes, scope/parity checks and independent paired replay reduce but do not eliminate implementation risks. All scored checker edits are automatic; old manual/mixed rows

are excluded. The native-only fixture guard rejects zero E1/E3 candidates, but is not an identical-policy or target-correctness guarantee. Selection probes overlap the evaluated cohort; reused executions remain labelled. Repeats do not create fresh-subject confirmation or isolate static/dynamic components. Configured temperature is not proof of transmitted sampling control.

Construct. Coverage, report burden and secondary PDS measure paired-version warnings, not independently adjudicated target findings; G08 is a counterexample and G24 leaves an ambiguity about suppression and target preservation. F_1 uses 78 version decisions, not deployment precision or target recall. A recovered warning may be noisy, notably G12. Scope includes patch-related objects rather than only the changed line, but does not certify semantics on other sites. The partial gate preserves nonzero vulnerable warning presence, not the entire warning set or every target location.

External. The 39-case one-per-commit cohort is not a random screen of all 286 upstream records or independent projects. TP/TN-based input selection may favor some artifacts; no population effect rate is claimed. The complete 15-case noisy stratum and 24 silent subjects have different tasks; KNighter does not attempt recovery without reports. E4 covers only the original 12-subject subset, omitting bounds categories, with one decode per configuration. Requested aliases do not identify immutable revisions. The CodeQL example is one patch-local case with inherited filters, not broad backend transfer. Perturbations informed the motivating edits and are not an unseen-code or future-version evaluation.

Conclusion. Repeated decodes are grouped by subject, not treated as independent bugs. Small non-random samples, prior configuration selection and stochastic outcomes limit uncertainty estimation. Native/control F_1 is nearly equal and no E3 subject benefits comparatively in all three decodes. We report all weaker/noisy outcomes and claim conditional observations, not statistically significant universal superiority.

9 Related Work

Static Analysis and Checker Languages. Static-analysis systems provide the substrate that SEMWEAVER refines against. Path-sensitive analyzers, abstract interpreters, and symbolic engines expose guards, states, paths, and diagnostics reusable as evidence [3, 6, 13, 15] and deploy at industrial scale [4, 7, 48]. Declarative languages offer a complementary approach by querying program facts through relational rules and points-to specifications [2, 5, 24], while semantic patching encodes recurring transformations over systems code [43]. SEMWEAVER introduces no new analyzer or query language. It treats LLM-generated analyzer artifacts as editable targets and uses analyzer-native facts as evidence for semantic completion.

Vulnerability Detection and Patch Evidence. Static vulnerability detection has long used taint and data-flow reasoning [1, 25, 36], graph and AST representations [55, 56], and learning-based token-, graph-, slice-, and line-level models [17, 34, 35, 46], though datasets and evaluations show that semantic robustness remains difficult [8, 16]. Clone- and patch-driven systems use historical fixes as compact witnesses [22, 28, 33]. These motivate our use of patches as specifications, but SEMWEAVER targets a different artifact: an already generated checker whose fixed-side behavior reveals missing semantics.

LLM-Based Code and Security Reasoning. Code LLMs synthesize programs [11], and agents such as RepoAudit audit repositories [19]. LLMDFA decomposes compilation-free dataflow analysis and uses synthesized tools for extraction and feasibility checks [54]. Generated artifacts still require security validation [44, 45]. SEMWEAVER edits a static-analysis artifact; paired analyzer executions, not the model’s explanation, determine acceptance.

Checker Synthesis and Feedback-Guided Repair. Patch-driven checker synthesis, especially KNighter [57], is the closest prior work. LLMs transform historical security fixes into specialized

analyzers that compile, execute, and are reused. SEMWEAVER is complementary because it assumes such a detector exists and refines it post-synthesis. A generator’s validation feedback reveals which side a detector fires on but not whether the missing predicate is a guard, barrier, flow, state, lifetime, type filter, or API-usage relation.

The loop also relates to repair, synthesis and counterexample-guided workflows using compiler/test/verifier feedback [12, 27, 30, 38, 39, 41, 42, 53], and iterative LLM feedback, tool use and search [40, 49, 52, 58, 59]. Here the edited artifact is a detector, not the vulnerable program. Paired execution retains warning recovery or report reduction with vulnerable warning presence, not compilation or suppression alone; this does not prove target detection.

Analyzer Context for LLMs and Alert Triage. LLIFT combines static analysis and LLMs for post-constraint-guided use-before-initialization reasoning [31]. Chapman et al. interleave intermediate results and prompts to infer error specifications [9]; IRIS infers taint specifications and analyzes CodeQL context [32]. LLM4PFA filters reports through interprocedural feasibility reasoning [14], and ZeroFalse enriches alerts before LLM adjudication [21]. VulGenie combines patch-dependency and attack/defense value-flow analysis with LLM-assisted API-rule synthesis and adaptive Java detection [10]. We claim neither first use of analyzer information nor new path-feasibility reasoning. Our operation completes an existing checker using auditable analyzer and checker context with paired warning outcomes, not a universal causal or precision claim. Evidence origin, selected edits, failed attempts and scope remain inspectable.

10 Conclusion

SEMWEAVER contributes an auditable post-synthesis checker editor with provenance-labelled compiler/checker context and paired validation. Across 39 generated-only CSA subjects, native GPT reaches 23 warning-positive vulnerable revisions and version-level $F_1 = 0.575$, versus KNIGHTER’s 15 and 0.448. In the complete 15-case noisy stratum, fixed reports fall from KNIGHTER’s 69 to 44 with all vulnerable warning signals retained, primarily through one subject. Overall native noise increases, F_1 nearly ties no-internal, and repeated benefits depend on sampling. A separate automatic CodeQL example and three-model subset provide bounded feasibility and sensitivity evidence. These are conditional checker-completion tradeoffs, not universal precision gains, target-recall proof or unseen-site generalization.

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In paper writing, the authors used generative AI tools (ChatGPT and Claude) solely for language polishing and clarity-oriented editing.

Data Availability

The anonymized repository is at <https://anonymous.4open.science/r/SemWeaver-FD1E/>. The English package contains implementation, frozen inputs, raw-output/provenance bindings, both systems’ outputs, original exchanges with English reading companions, all reported repeated/model-subset results, and offline byte/reaggregation checks. Historical manual/mixed-provenance materials are unscored. Full reruns require public Linux revisions and the pinned toolchain; the package is intended to remain public upon acceptance.

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